Faculty of Sciences

Department of Computer Sciences

Master thesis

Procedural Generation of Solvable Levels in Multi-Agent Reinforcement Learning Environment

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Introduction

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Methods

4.1 Solver by Reduction to SAT

4.1.1 Definition of Sets

The following sets are used to represent the domain of the problem.

- C: colors of agents, $C \in \{\{1\}, \{1,2\}, \{1,2,3\}, \{1,2,3,4\}\}$ depending on the number of agents in the level
- D: direction of lasers $\{N, S, E, W\}$
- P: all possible positions for the level in a coordinate (x, y)
- T: all time steps from 0

4.1.2 Definition of Variables

In this section, we define all propositional variables used in the SAT encoding.

- $a_{c,x,y,t}$: agent with color $c \in C$ at position (x, y) after t time steps
- $b_{c,d,x,y,t}$: beam of color $c \in C$ with direction $d \in D$ at position (x, y) after t time steps
- $l_{c,x,y,t}$: laser of color $c \in C$ active at position (x, y) after t time steps

4.1.3 Constraints and Logical Encoding

We now formalize the constraints ensuring the correctness of the reduction. Each group of clauses corresponds to one logical component of the problem.

1. Initialization:

   - Agents:
     The initial position of the agents is fixed by the level design. We also say where it can so there is no duplication of agent.

$$\bigwedge_{c \in C} \bigwedge_{(x,y) \in P} \begin{cases} a_{c,x,y,0} & \text{if there is a starting tile} \\ \neg a_{c,x,y,0} & \text{else} \end{cases}$$

- Lasers:
  We define a beam always active at the source of the laser, and then I will propagate it with the rules of the laser activity.

$$\bigwedge_{c\in C}\bigwedge_{d\in D}\bigwedge_{(x,y)\in P}\bigwedge_{t\in T} b_{c,d,x,y,t} \qquad \text{if there is a laser source}$$

2. Agents movements

- Legal movement:
  If there is an agent at position (x, y) at time t, then there must be an agent at position (x, y) or one of its neighbors at time $t + 1$ (forward consistency).

$$\bigwedge_{c\in C}\bigwedge_{t\in T}\bigwedge_{(x,y)\in P} a_{c,x,y,t} \rightarrow \bigvee_{(x',y')\in \text{ next}(x,y)} a_{c,x',y',t+1}$$

$$\Updownarrow$$

$$\bigwedge_{c\in C}\bigwedge_{t\in T}\bigwedge_{(x,y)\in P} \neg a_{c,x,y,t} \vee \bigvee_{(x',y')\in \text{ next}(x,y)} a_{c,x',y',t+1}$$

  where $\text{next}(x, y) = \{(x', y') \in \{(x, y), (x - 1, y), (x + 1, y), (x, y - 1), (x, y + 1)\} \mid \text{If the position } (x', y') \text{ is in the grid and there is no wall or laser source}\}$

Two different method to constraint the duplication of agents, so an agent cannot be in two different positions at the same time.

- Global Method:
  There is at most one agent of color c at time t.

$$\bigwedge_{c\in C}\bigwedge_{t\in T}\bigwedge_{\substack{(x_1,y_1),(x_2,y_2)\in P\\(x_1,y_1)\neq(x_2,y_2)}} \neg a_{c,x_1,y_1,n} \vee \neg a_{c,x_2,y_2,n}$$

- Local Method:
  At most one agent per time step between next positions.

$$\bigwedge_{c\in C}\bigwedge_{t\in T}\bigwedge_{\substack{(x',y'),(x'',y'')\in \text{ next}(x,y)\\(x',y')\neq(x'',y'')}} \neg a_{c,x',y',n} \vee \neg a_{c,x'',y'',n}$$

  Agents should come from a legal movement, so if there is an agent at position (x, y) at time $t + 1$, then there must be an agent at position (x, y) or one of its neighbors at time t (backward consistency).

$$\bigwedge_{c\in C}\bigwedge_{t\in T}\bigwedge_{(x,y)\in P} a_{c,x,y,t+1} \rightarrow \bigvee_{(x',y')\in \text{ next}(x,y)} a_{c,x',y',t}$$

$$\Updownarrow$$

$$\bigwedge_{c\in C}\bigwedge_{t\in T}\bigwedge_{(x,y)\in P} \neg a_{c,x,y,t+1} \vee \bigvee_{(x',y')\in \text{ next}(x,y)} a_{c,x',y',t}$$

- Overlapping agents:
  Two different agents cannot be on the same position at the same time and can not move on an occupied position.

$$\bigwedge_{\substack{c_1,c_2\in C\\ c_1\neq c_2}} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} \neg a_{c_1,x,y,t} \vee \neg a_{c_2,x,y,t}$$

$$\bigwedge_{\substack{c_1,c_2\in C\\ c_1\neq c_2}} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} \neg a_{c_1,x,y,t+1} \vee \neg a_{c_2,x,y,t}$$

$$\bigwedge_{\substack{c_1,c_2\in C\\ c_1\neq c_2}} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} \neg a_{c_1,x,y,t} \vee \neg a_{c_2,x,y,t+1}$$

- Agents must be on exit to win:

$$\bigwedge_{(x,y)\in P} \bigvee_{c\in C} a_{c,x,y,t_{\text{MAX}}} \quad \text{if there is an exit tile at position } (x,y)$$

3. Lasers activity

- A beam can not propagates if there is a wall:

$$\bigwedge_{c\in C} \bigwedge_{d\in D} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} \neg b_{c,d,x,y,t} \quad \text{if there is a wall at position } (x,y)$$

- A beam propagates if and only if there is no agent to block it:

$$\bigwedge_{c\in C} \bigwedge_{d\in D} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} \big(b_{c,d,x,y,t} \wedge \neg a_{c,x',y',t}\big) \leftrightarrow b_{c,d,x',y',t}$$

$$\Updownarrow$$

$$\bigwedge_{c\in C} \bigwedge_{d\in D} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} \big(\neg\big(b_{c,d,x,y,t} \wedge \neg a_{c,x',y',t}\big) \vee b_{c,d,x',y',t}\big) \wedge \big(\big(b_{c,d,x,y,t} \wedge \neg a_{c,x',y',t}\big) \vee \neg b_{c,d,x',y',t}\big)$$

$$\Updownarrow$$

$$\bigwedge_{c\in C} \bigwedge_{d\in D} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} \big(\neg b_{c,d,x,y,t} \vee a_{c,x',y',t} \vee b_{c,d,x',y',t}\big) \wedge \big(b_{c,d,x,y,t} \vee \neg b_{c,d,x',y',t}\big) \wedge \big(\neg a_{c,x',y',t} \vee \neg b_{c,d,x',y',t}\big)$$

$$\text{where } (x',y') = \begin{cases} (x,y-1) \text{ if } d=N \\ (x+1,y) \text{ if } d=E \\ (x,y+1) \text{ if } d=S \\ (x-1,y) \text{ if } d=W \end{cases}$$

- Link value of laser and beam:

$$\bigwedge_{c\in C} \bigwedge_{d\in D} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} b_{c,d,x,y,t} \leftrightarrow l_{c,x,y,t}$$

$$\Updownarrow$$

$$\bigwedge_{c\in C} \bigwedge_{d\in D} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} \big(b_{c,d,x,y,t} \rightarrow l_{c,x,y,t}\big) \wedge \big(l_{c,x,y,t} \rightarrow b_{c,d,x,y,t}\big)$$

$$\Updownarrow$$

$$\bigwedge_{c\in C} \bigwedge_{d\in D} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} \big(\neg b_{c,d,x,y,t} \vee l_{c,x,y,t}\big) \wedge \big(\neg l_{c,x,y,t} \vee b_{c,d,x,y,t}\big)$$

- Agents can not step on active laser:

$$\bigwedge_{\substack{c_1,c_2 \in C \\ c_1 \neq c_2}} \bigwedge_{(x,y) \in P} \bigwedge_{t \in T} l_{c,x,y,t} \rightarrow \neg a_{c,x,y,t}$$

$$\Updownarrow$$

$$\bigwedge_{\substack{c_1,c_2 \in C \\ c_1 \neq c_2}} \bigwedge_{(x,y) \in P} \bigwedge_{t \in T} \neg l_{c,x,y,t} \vee \neg a_{c,x,y,t}$$

Chapter 5

Experiments

Conclusion

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